



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB XYZ Panner

Three-Axis Spatial Panner

Reliable stereo panning with added cues for height, distance, and front-to-back position, rendered for speakers or for headphones with KEMAR HRTF.

Top and side views

Stereo or KEMAR HRTF headphones

3 distance curves

Surround-aware output



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01 Overview

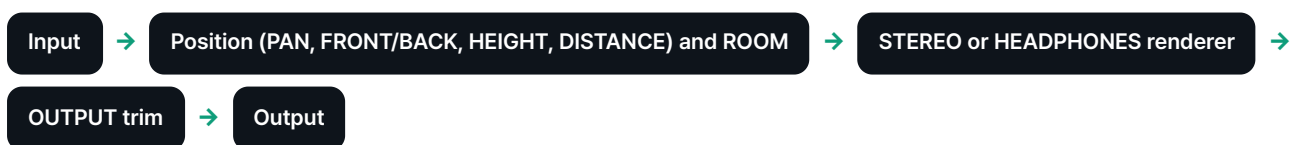
TB XYZ Panner keeps dependable stereo panning at its core and adds procedural cues for height, distance, and front-to-back perspective.

The top view places a source left, right, in front, or behind without changing its distance. The side view shapes elevation and depth through level, tone, and early-reflection behaviour. Stereo mode gives solid conventional placement; Headphones mode uses KEMAR HRTF rendering for a more immersive binaural image. ROOM and OUTPUT finish the space.

Key features

- Top view for left/right and front/back, side view for height and distance
- **PAN**, **FRONT/BACK**, **HEIGHT**, **DISTANCE**, and **ROOM** controls
- Three distance curves: Physical, Near Field, and Linear
- STEREO rendering for speakers, HEADPHONES rendering with KEMAR HRTF
- On surround outputs, the non-LFE channels are mapped by speaker direction
- **OUTPUT** trim, undo, and redo

Signal flow



About this manual

This manual describes TB XYZ Panner 1.3.0 for Windows (VST3). The current macOS build is version 1.0.2 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB XYZ Panner on a mono or stereo track. By default the source is centred, in front, at ear height, close, with **ROOM** at 35%.
- 2 Choose **STEREO** 1 for speakers or **HEADPHONES** for binaural listening.
- 3 Place the source left or right and front or back in the top view 3, or with **PAN** and **FRONT/BACK** 5.

- 4 Set height and distance in the side view 4, or with HEIGHT and DISTANCE.
- 5 Add space with ROOM 6 and set the level with OUTPUT 7.

03 Interface



Captured from version 1.2. Version 1.3.0 adds a DISTANCE CURVE selector (PHYSICAL, NEAR FIELD, LINEAR); the other controls are the same.

- 1 **STEREO / HEADPHONES** — the render mode.
- 2 **Undo and Redo** — step back and forward through changes.
- 3 **Top view** — left, right, front, and back around the listener.
- 4 **Side view** — up and down, and near to far.
- 5 **POSITION** — PAN, FRONT/BACK, and HEIGHT.
- 6 **SPACE** — DISTANCE and ROOM.
- 7 **LEVEL** — OUTPUT trim.

04 Controls

Position

Control	Range · default	What it does
PAN	Left 100% to right 100% · 0% (centre)	Left-right position.
FRONT/BACK	Back 100% to front 100% · front 100%	Front-back position. The panel shows F or B with the amount.

Control	Range · default	What it does
HEIGHT	Down 100% to up 100% · 0%	Elevation, shaped through level, tone, and early-reflection cues.

Space and level

Control	Range · default	What it does
DISTANCE	0 to 100% · 0%	How far the source is from the listener. Distance changes level, tone, and reflections.
DISTANCE CURVE	PHYSICAL, NEAR FIELD, LINEAR · PHYSICAL	How the level falls as DISTANCE increases. Physical is the default.
ROOM	0 to 100% · 35%	Amount of room reflections around the source.
OUTPUT	-24 to 0 dB · 0 dB	Gain after rendering. Use it to keep headroom.

Render mode

Mode	What it does
STEREO	Conventional speaker placement. On surround tracks such as 5.1 or 7.1, the non-LFE channels are mapped by speaker direction. Surround behaviour can depend on the host.
HEADPHONES	Binaural rendering with KEMAR HRTF for headphone listening. The result depends on your headphones and the shape of your head.

Check a mix made in Headphones mode on speakers too, and the other way round. HRTF cues that work on headphones can sound coloured on speakers.

05 Recipes

Starting points for common jobs. Adjust them by ear.

Push an element back

Make a pad or backing vocal sit behind the lead.

DISTANCE	40 to 70%
ROOM	35 to 60%
DISTANCE CURVE	Physical

- 1 Raise DISTANCE until the element sits behind the lead, then adjust ROOM for depth.

Overhead effect on headphones

A sound that seems to come from above.

Render mode	HEADPHONES
HEIGHT	60 to 100%

- 1 Keep the source in front at first; height cues are clearer from the front.

Behind the listener

Ambience or effects that come from behind on headphones.

Render mode HEADPHONES

FRONT/BACK Toward back

- 1 Front-back cues are subtle on speakers; check them on headphones.

Even, predictable fades

Distance automation with a simple level curve.

DISTANCE CURVE Linear

Automation Distance

- 1 Automate DISTANCE and listen to the level change across the move.

06 Tips and troubleshooting

Symptom	What to check
The sound is too quiet when far away.	Distance lowers the level. Try another DISTANCE CURVE, or raise OUTPUT up to 0 dB.
Height or front-back is hard to hear on speakers.	These cues are strongest in HEADPHONES mode.
Headphones mode sounds coloured.	HRTF filtering changes the tone. Use STEREO for speaker mixes.
The output clips.	Lower OUTPUT to keep headroom.
Surround channels behave unexpectedly.	Check the track's channel layout in the host; support for surround layouts can vary by host.

07 Parameter reference

Panel label	Host parameter	Range	Default
PAN	Pan	-1 (left) to +1 (right)	0
FRONT/BACK	Front Back	-1 (back) to +1 (front)	+1
HEIGHT	Height	-1 (down) to +1 (up)	0
DISTANCE	Distance	0 to 1	0
ROOM	Room	0 to 1	0.35
DISTANCE CURVE	Distance Model	Physical, Near Field, Linear	Physical
STEREO / HEADPHONES	Render Mode	Stereo, Headphones	Stereo
OUTPUT	Output Trim	-24 to 0 dB	0 dB

08 Version history

Version	Changes
1.3.0 · 2026-09-30	Added the Physical, Near Field, and Linear distance attenuation curves, with Physical as the default.
1.2.0 · 2026-07-30	Moved the effect logo and product name into a shared header.

09 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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